

Sleeping Machines: the investment case

Sleeping Machines aims to be a **universal trainable computing substrate**: one composable architecture for language and reasoning, multimodal world models, embodiment, typed tabular data, synchronous samples, asynchronous events, anonymous interleaved process traces, token sequences, continual learning, communication, self-design and hardware.

The ambition combines distinct design axes: **computation through sleeps and temporal races; globally clockless event-driven execution; learning in the same temporal, addressed and asynchronous substrate as inference; capacity beyond selected activity; and tunable work for small-data, low-resource and large-scale settings**. Data modality is a separate axis from each of these. Scaling advantages and economical intelligence from sensors to datacenters are targets to measure. See [the full vision](#).

A learning architecture and computing substrate for intelligence everywhere.

Prepared 3 October; revised 5 October 2026. Investment thesis and proposed commercial strategy; completed research evidence is identified separately from product ambitions.

Current terms: **€3M raise at €50M priced pre-money**, with €100M as a separate stretch scenario. Read the [current valuation rationale](#) and [application opportunity register](#). The [previous memo](#) retains the earlier \$10M discussion for historical review.

The proposition

Sleeping Machines aims to make capable intelligence economical wherever it is needed: in datacenters, on personal devices and inside machines that interact with the world. We are developing models and their execution substrate together. Time performs computation; messages race to select useful work; persistent addressed memories retain context; counterfactual credit teaches alternatives that did not win. The long-term implementation target is a globally clockless, event-driven substrate supporting both inference and learning.

The investment opportunity is ownership of a useful new way to build, train and execute AI. A successful platform could earn revenue through model/runtime licensing, deployment software, accelerator IP and eventually hardware. Each route draws on a common architecture and toolchain. The initial product must solve a specific customer problem; the broader platform is the expansion path.

The ambition is the next general-purpose substrate for machine intelligence: one architecture, learning rule and execution model spanning frontier language and reasoning, multimodal world models, embodied intelligence and robotics, event-native analytics, continual on-device learning, adaptive communication and self-designing models, on hardware from datacenters to phones, robots, sensors and clockless event processors that learn on chip. General intelligence is the overarching aspiration. The case rests on a working research program with concrete positive results and a staged way to turn them into customer value; frontier capability, commercial serving savings and clockless chips are the milestones this round funds.

Why this could become a foundational platform

AI needs capability under constraints. A datacenter buyer cares about useful output per dollar, rack watt and unit of resident memory. A mobile buyer cares about capability under battery, heat, storage and connectivity limits. A robot needs timely decisions, persistent context and adaptation as its environment changes. These are different commercial requirements, with a shared technical question: how much useful intelligence can a system obtain from the work it actually performs?

Our architecture explicitly separates available capacity, scored keys, selected state updates, delivered values and counterfactual learning work. This makes capacity beyond selected activity a design objective. Deep memories and learned routes allocate work to the current need while preserving a larger store of skills and context. Scalable key discovery is the next engineering step (the current implementation scores all keys in its pools).

If these mechanisms improve the quality/resource frontier, the benefit can be spent on lower cost or greater capability at the same budget. For example, a demonstrated 40% reduction in a relevant total cost would permit about 1.67 times the corresponding work at the same budget. The scaling study measures whether the advantage widens with scale.

The advantage portfolio

Mechanism	Potential customer advantage	Present evidence and next boundary
Time as computation and race attention	Select and combine information through learned delays and arrival order; match irregular streams	Temporal/race implementations and integrated fits exist; precision, jitter and physical timing need hardware validation
Sparse addressed updates and separate keys/values	Deliver and update useful content without computing every candidate value during inference	Winner-only evaluator and arithmetic ledgers exist; actual-trained parity, rescore and total runtime remain pending
Persistent deep event representations	Maintain context across observations and compute from incoming messages plus stored state	Integrated language/event models learn; latest language scores use segment resets, not demonstrated indefinite memory
Capacity beyond selected activity	More useful state or skills at bounded selected work	Doubling p32 receiver slots improves quality at eight writes per input; key scoring, learning and storage grow
Counterfactual route credit	Train sparse hard choices and deepen useful computation without dense inference	Alternative-value credit improves completed depth-4 and depth-8 language fits; training alternatives still cost work
Event-driven, globally clockless, memory-local hardware	Energy from sparse local memory access (data movement dominates AI-chip energy) plus no clock tree and leakage-only idle capacity; average-case latency; thermal headroom for memory-on-logic stacking; a competitive model for the existing neuromorphic chip class (hardware thesis)	Mechanisms and energy accounting derived; no fabricated Sleeping Machines chip or measured joule advantage; local on-chip learning rule open
Online learning	Adapt to users, environments and drift near the point of use	Causal full-backbone CPU pilot improves predictions; stable continual learning and on-chip optimizer execution remain open
Computation during silence	Preserve relevant temporal evolution without periodic scans; reason about absence when needed	Temporal algebra and silence-aware supervision are part of the research; deadlines, readouts and physical retention are paid
Models and substrate developed together	Align learning, memory, execution and physical implementation rather than optimize one cost in isolation	Shared theory, implementations, contracts and audited results exist; an integrated commercial stack is still to be delivered

Clockless circuits, sparse models, recurrent state and on-chip learning have precedents. Intel's Loihi 2 describes asynchronous cores and programmable learning. Our differentiation is the complete trainable construction: deep content-bearing temporal computation, addressed persistent memory and credit to unrealized hard routes, translated into a useful system. [Intel Loihi 2](#)

What already makes the case tangible

Public leaderboard wins on the family's home field (6-7 October 2026). On EasyTPP (ICLR 2024), the standard benchmark for marked event streams in continuous time, the model is a race of delayed clocks over persistent temporal memory. Official splits, the published per-event log-likelihood protocol, 5 seeds, sealed test scored once per seed ([dossier](#)):

Dataset (nats/event)	Best published	Ours	Compute vs S2P2
Taobao	1.318 (IFTTP)	1.399 ± 0.003	0.92×
Taxi	0.522 (S2P2)	0.525 ± 0.001	1/12
Taxi, 5-seed mixture	0.522 (S2P2)	0.536	0.41×
StackOverflow	-2.163 (S2P2)	-2.144 ± 0.004	1.26×
StackOverflow, matched	-2.163 (S2P2)	-2.153 ± 0.005	1.015×
Retweet	-6.348 (NHP)	-6.326 ± 0.001	1/15
Amazon	0.781 (S2P2)	0.803 ± 0.001	0.29×

Higher is better (log-likelihood, nats per event); compute is per-event inference work relative to S2P2. Every seed is ahead of the best published result on every dataset.

A second public win, with the same core, in clinical early warning (7 October 2026). On P19 (PhysioNet 2019 sepsis prediction, 38,803 ICU stays, 34 irregularly sampled channels, 4.2% positive), five official splits, TEST scored once per split: **AUROC 0.916 ± 0.022 and AUPRC 0.639 ± 0.039** against the best published 0.903 / 0.583 (MTM, 2025), with 62,681 parameters ([dossier](#)). The classifier reuses the EasyTPP model's temporal memory layer — same code, same size — adding addressed channel memories with sufficient statistics, typed comparisons and time-since-measurement as information. Paired five-split tests (8 Oct): removing the model's temporal memory (same network reading only its statistic slots) drops it to 0.900 / 0.572 on every split (paired Δ AUPRC +0.072, $t = 31$), and gradient-boosted trees on the same statistics score 0.914 / 0.618: the temporal memory carries signal the statistics lack, and the model leads the trees on AUPRC on every split (+0.026, paired $t = 4.2$; AUROC level). The result reproduces from scratch on separate hardware (AUROC within 0.0004 on all five splits).

A third public win, in wearable sensors (8 October 2026). On PAM (activity recognition from wearable inertial sensors, eight classes), five official splits, TEST scored once per split under a pre-registered protocol: **accuracy 0.978 ± 0.007 and F1 0.980 ± 0.008** against the best published 0.975 / 0.976 (MTM, 2025), with 46,316 parameters. Four of five splits are ahead; split 4 scores 0.964.

Generality, stated precisely. The EasyTPP leader S2P2 (NeurIPS 2025, GE HealthCare and UC Irvine) is shown on one task type. One Sleeping Machines family now wins public benchmarks on generative event modelling and on sparse clinical classification (P19) and wearable sensors (PAM), with further evidence on process logs, tokenized and character language and few-shot temporal reasoning; large-scale language and real tables remain behind. One shared temporal core (14,080 parameters) trained jointly on Taxi, Taobao, StackOverflow and Amazon matches or beats the separate per-dataset models on three of four (development data): mobility, shopping and Q&A timing share one learned dynamics. Generative pretraining, on the same clinical cohort or on 31K stays from another, gave no robust label-efficiency gain on mortality prediction (mean +0.004 AUROC at 10% labels) ([generality plan](#)).

- **Taobao and Taxi are confirmed wins** at a fraction of the state of the art's compute.

- **StackOverflow is a confirmed win** in accuracy (+0.019 nats/event) at 1.26× compute, from a continuous-time state clock that passed the recording-grid audit.

- **Retweet is a confirmed win:** -6.326 ± 0.001 vs the best published -6.348 (NHP), all five seeds ahead, at 1/15 of S2P2's parameters and per-event compute.

- **Amazon is a confirmed win (7 October):** 0.803 ± 0.001 vs S2P2 0.781 ± 0.011 , all five seeds ahead, with the best published time and type log-likelihoods, at 0.29x S2P2's per-event compute and 3.6x fewer parameters.

- **All five EasyTPP datasets are won**, against a pass criterion of two.

- **How Amazon was won:** the previous protocol (0.784 ± 0.027) had one seed stuck in a weaker optimization basin; the diagnosis (delay windows drifting beyond all observed gaps) led to windows anchored to their gap component, which removed the basin: five seeds within 0.0015 of each other, one run each, no restarts.

- These are the first public-leaderboard results of the family. They show the core primitive is competitive on public terms where its mathematics fits: a race of clocks *is* a temporal point process.

These completed language comparisons use text8, 10M fitting characters, one pass and the saved controls' T256 evaluation window. Each native row is a single seed. Fitting work includes backward, route-credit and optimizer estimates. Inference work is per evaluated input position; native winner-only values are shape traces pending actual-trained backend admission, not measured serving.

Model	Test bpc, lower better	Whole fitting TFLOPs est.	Fit MFLOPs/input position est.	Inference MFLOPs/input position est.
Native p32/D4, timing credit	2.5064	7.22	0.72	0.163
Native p32/D4, alternative-value credit	2.3715	7.24	0.72	0.163
Native p32/D4/pool4, alternative-value credit	2.3452	11.95	1.20	0.164
Native p64/D4, alternative-value credit	2.1833	26.79	2.68	0.605
Native p96/D4, alternative-value credit	2.1625	58.65	5.87	1.324
Saved LSTM-256	2.1706	20.31	2.03	0.677
Saved Transformer-256x2	2.4269	111.26	11.13	3.710

Three observations are particularly relevant to investment:

- **A learning repair unlocks meaningful capability.** Value-informed route credit improves p32/D4 by 0.135 bpc with about 0.3% more counted fitting work and unchanged hard forward behavior. The improvement also appears in a completed depth-8 model. This is evidence that learning the alternatives matters, rather than evidence that additional activity alone explains gains.

- **The sparse native construction beats Transformers at lower compute.** Credited p32/D4 beats the saved one-pass Transformer by 0.0554 bpc with about 15.4 times less estimated fitting work and 15.2 times fewer parameters. With more passes, the native model reaches 1.888 bpc against the 4-pass Transformer-256x4's 1.908 at 0.40x its training and 0.18x its inference compute.

- **Quality and useful capacity improve.** The p96 model slightly exceeds the saved LSTM's quality, while using about 2.9 times its fitting work. At p32, doubling slots from 16 to 32 improves 2.3715 to 2.3452 with eight selected writes unchanged, but 16 to 32 scored keys and about 1.65 times fitting work. Useful capacity grows beyond selected activity.

- **Tuned baselines at our budgets (5 October 2026).** Ten dense baselines retuned at the native compute budgets and selected by validation: the native model beats every tuned Transformer at both budgets (1.888 vs 1.996; 1.955 vs 2.215 bpc, at about equal compute). Tuned small LSTMs lead at

these budgets (1.825 at ≤ 352 TF; 1.915 at ≤ 107 TF), on local 2–4-character modelling. The strategic contest is against Transformers at scale.

A separate integrated full-backbone online CPU experiment scores 3.1909 bpc with frozen weights versus 3.0957 with adaptation on 8,191 new development targets. Predictions precede each 16-character block update; both arms have persistent event state. Next: gains under drift with retained quality and bounded update cost.

The program also spans event speech, event vision, tabular tasks, interleaved industrial event logs and temporal prediction, each at its own maturity; the report lists every win and loss per domain. Invalid target-dependent mixtures are quarantined and excluded from this investment case.

Completed parents and exact values are indexed by the accompanying evidence manifest. [Research status](#)

The numerical table above remains the frozen 3 October 10M comparison. At 90M characters, against tuned references at equal training compute, the results are mixed:

- $\approx$ **0.96 PF**: native 1.800 bpc loses to a tuned LSTM-512 (1.729) and a tuned Transformer-192 $\times$ 4 (1.780).

- $\approx$ **2.1 PF**: native 1.783 beats a tuned Transformer-256 $\times$ 4 (1.811) at 1.06 $\times$ its compute, while a Transformer-192 $\times$ 4 (1.704) leads.

The gap to the best dense model widens from 10M to 90M; that trend is the central language problem (report Part I §4.3).

Tokenized language (8 October 2026). A temporal-memory token model with the keyed predecessor read scores 6.009 ± 0.010 nats per token (3 seeds) on the 65,528-target GPT-2-tokenized FineWeb slice at 1M training tokens, against a Kneser–Ney trigram's 6.537 and the trigram at 4 $\times$ the data (6.100); at 4M tokens 5.521 vs 6.100, so the lead grows with data. The keyed predecessor read behind it also solves associative recall across irregular gaps (97.5%, 3 seeds) (report Part I §4.5). The count models are a floor, not the frontier; a published small Transformer under the same protocol is the next reference. See the [current evidence map](#).

Expanded platform opportunities

The [opportunity register](#) specifies irregular-stream ingestion, asynchronous neural codecs, task-oriented communication, learned memory/update policies and joint embodied cognition. These can exploit one event/state/learning interface across local and remote regions. Variable-rate coding, irregular-input models and joint VLA learning already have precedents. The proposed value is their economical temporal and selective integration, with first proof conditions and current gaps explicit.

Joint motor/reasoning learning can teach shared physical abstractions and planning skills; transfer in each direction is tested separately on held-out tasks. General intelligence is the overarching aspiration.

Markets: one foundation, distinct products

Datacenters and frontier-model developers. Start with a reproducible quality/cost improvement on a defined workload. Buyers could license a model, runtime or accelerator design that increases useful output within their power, memory and latency budgets. Efficient inference is one entry; efficient training and stronger scalable models are larger opportunities. A software proof can precede the capital demands of custom silicon.

The IEA's updated outlook puts datacenter electricity at 485 TWh in 2025 and projects 950 TWh in 2030. That is the scale of the constraint (not all of it is addressable AI spend); validated quality per watt and dollar has substantial economic value. [IEA outlook](#)

Mobile, wearables and personal devices. The desired product is an SDK/model and later licensed compute IP for continuously available contextual AI: speech, sensing, assistance and personalization under a device power budget. Persistent state and local learning could reduce cloud dependence and feedback latency. Keeping processing local can reduce raw-data transfer, but the product must implement its privacy and update policy. Online learning must be shown to improve usefulness without unacceptable forgetting or additional energy.

GSMA's 2026 report describes 5.8 billion unique mobile subscribers and 8.8 billion wireless connections. Those are ecosystem counts, not compatible device shipments or paying customers. The opportunity is an OEM deployment that becomes a repeatable design win, then expands across device families. [GSMA Mobile Economy 2026](#)

Robotics and autonomous machines. Continuous sensor fusion, memory, low-latency response and adaptation offer a natural application for an event substrate. The proposed product is a model/runtime or embedded accelerator that improves a measured perception or adaptive-control workload. Learning from deployment could eventually address changing payloads, environments and sensor characteristics. Closed-loop transfer, latency, robustness and retained skills are proven in robotic trials, the next stage after recorded-stream models.

IFR reports five million industrial robots operating in 2025 and more than 600,000 new installations that year. This is a concrete industrial ecosystem, not a forecast of our sales; service robots and future autonomous devices are additional possibilities. [IFR World Robotics 2026](#)

Everything between edge and cloud. Industrial monitoring, network/telecom streams, connected vehicles, private enterprise AI, environmental sensing and adaptive forecasting share aspects of the same opportunity. Some need sparse continuous sensing; others need capable private inference or adaptation. They are expansion options after a repeatable first product, with their own data, quality and deployment protocols.

Define the upside in economic terms

The following are transparent, independent scenarios, not forecasts, observed prices or a computed market size. They show how a narrow deployment can become a venture-scale business without requiring an immediate win in every domain.

Route	Explicit hypothetical assumptions	Annual company revenue implied
Datcenter software/IP	Reach \$1B/year of eligible customer execution spend; reduce total eligible cost 20%; capture 10-20% of the \$200M value created	\$20-40M
Mobile compute IP	Win 100M newly licensed devices/year at \$0.25-\$1.00 per device	\$25-100M
Robotics/industrial runtime	Serve 1M paying active units at \$20-\$100 per unit/year	\$20-100M
Scaled platform	Achieve \$250M recurring annual revenue across a validated product portfolio	\$250M; requires adoption and sustained delivery

The datacenter scenario's \$200M saving precedes our fee: a \$20-40M fee leaves \$160-180M for customers before migration, support and update costs. Final customer savings must include all those costs. Device royalties are paid on actual licensed units; existing installed devices are not annual shipments. Robotics revenue assumes a recurring product, not an upfront chip sale. The routes can overlap, so do not sum them into a TAM. Chip revenue, margins and financing needs differ substantially from software/IP.

For scale intuition only, \$250M recurring revenue at an assumed 8-15 times revenue valuation would imply \$2.0-3.75B enterprise value. Neither that revenue nor those multiples is a prediction or a current comparable. A frontier-model and substrate platform with durable billion-dollar annual revenue could support much larger, potentially tens-of-billions outcomes under suitable economics. The route to that

upside is owning a widely adopted layer of AI infrastructure, with defensible quality/cost benefits and meaningful value capture.

This is why the upside can justify early investment: capital today buys an opportunity to establish that layer before the complete platform is proven. The opportunity has correlated technical risks across its products; several applications do not constitute independent chances of success.

Defensibility and capital efficiency

The moat combines a learning method, an execution contract, model/runtime engineering, memory/routing design, hardware mappings and a growing corpus of reproducible evidence. Successful deployments can add integration know-how, task-specific learning policies and developer adoption. The combination is harder to reproduce than any isolated primitive.

No patents are filed yet. The historical manifesto is public and cites a 2021 origin; the round funds a documented inventory of ownership, contributor rights, dependencies, licenses and protectable implementations, and a freedom-to-operate review.

Funding should convert this proposed moat into documented company assets: ownership and disclosure review, patent counsel's assessment, selective priority filings and trade-secret practice. These sit within the existing €200,000 legal/IP/operations allocation, subject to quotes and staged filing decisions. The founder's earlier patents belong to his former employer and contribute experience, not venture-owned rights. See the [IP protection plan](#).

There are financing precedents for both parts of the thesis: Liquid AI announced a \$250M Series A for efficient general-purpose models in December 2024; Innatera reported a EUR15M Series A for neuromorphic edge technology in March 2024. They demonstrate investor interest in these categories. Their teams, maturity and funding amounts do not establish our valuation. [Liquid AI announcement](#), [Innatera announcement](#)

The proposed sequence is capital-efficient: establish the algorithm and software resource frontier; secure a workload/design partner; validate fixed-precision hardware semantics on FPGA; fund ASIC development after quality, economics and demand justify it. A clocked FPGA can validate event semantics; it cannot demonstrate the energy of a fabricated clockless ASIC.

A basis for today's valuation

The current opening proposal is **€3M at €50M priced pre-money; €100M** is a stretch scenario. This prices the potential of a model/learning/runtime and computing-substrate platform, supported by research execution and a staged program. It is a negotiating position, not an independent appraisal.

Public de-risking since the proposal was set. The confirmed EasyTPP wins (Taobao, Taxi, StackOverflow) and the P19 clinical win are the first instances of the evidence the rationale names as most valuable: a repeated useful-quality and full-cost advantage, on a public leaderboard with published baselines.

- They make the €50M opening price materially more defensible.
- The €100M case needs the advantage beyond one benchmark family. Its three conditions hold as of 7 October (VALUATION_RATIONALE.md, condition status):
- further datasets at matched compute: StackOverflow at 1.015× S2P2's per-event compute;
- a second domain: P19 sepsis, AUROC 0.916 vs 0.903 published;
- an independent reproduction: the Taxi win rerun from scratch on separate hardware, 0.5252 ± 0.0007 vs 0.5250 ± 0.0010 (AWS) and S2P2's 0.522 (same team; a third-party rerun is next).

The rationale now treats asynchronous ingestion/output, learned memory policies and embodied/cognitive integration as concrete expansion options. The common structure could permit

end-to-end learning across their information and credit paths. Each has its prior art and first discriminating test.

Execution supports taking the ambition seriously: implemented integrated models, completed mechanism comparisons, larger-data learning, retained failures and protocol corrections, and reviewable source/resource contracts. Tero's relevant public engineering/research/invention history adds an execution case, subject to references and commitment. Scaling a company, building a team and establishing usable rights are separate obligations. Research co-authorship remains credited; sole-founder status does not settle ownership.

At €50M pre-money, €3M produces €53M post-money and 5.6604% initial investor ownership; the €100M case produces €103M and 2.9126%, before other dilution. The planned €3M/18-month program adds learning/runtime/evaluation capacity and stages compute and hardware feasibility against explicit gates. Software and one useful workload can create value before the full platform is complete.

The [valuation rationale](#) documents positive assets, execution evidence, financing precedents, equity arithmetic and remaining proofs. Comparable seed funding announcements establish category investability; unreported pre-money prices cannot be inferred from round amounts. Several applications share risks and must not be summed into a TAM or independent success probabilities. The vision supports exceptional upside; investor demand, rights, execution and reproducible benefits determine attainable terms.

What the next investment should buy

1. **A reproducible model advantage at scale.** Race attention against competent Transformers at equal complete compute on modern language data, across a scaling curve on GPU, with replications.
2. **Extend the home-field wins on event streams.** Public EasyTPP wins are achieved (Taxi, Taobao, StackOverflow). Next:
 - the remaining datasets at matched compute;
 - a second public domain (irregular clinical and sensor series: P12, P19, PAM);
 - the sealed FAS v2 confirmation on anonymous interleaved logs, against small LSTM/Transformer references;
 - identity-stripped real logs with a design partner.
3. **A deployable sparse backend.** Admit actual-trained winner/state/cache/RNG parity and heldout rescore, then measure setup, residency, latency, throughput and total cost. The prepared packed-weight worker is an initial runtime implementation; its lifecycle checks pass, native admission is open.
4. **One customer workload and one adoption route.** Define the buyer, useful quality level, deployment constraints and acceptable integration cost. Proposed first route: a software/model/runtime proof for datacenter AI, while validating event-stream strengths for later edge/robotics products.
5. **A credible adaptation result.** Demonstrate prequential gains under drift, retained stationary quality and bounded update resources against competent online baselines. Distinguish state updates from learned weight adaptation.
6. **A hardware execution contract.** Export validated traces; establish event ordering, precision, timing, local state and backpressure. Measure clock, communication, memory and learning overhead before an ASIC commitment.
7. **Company-owned IP protection.** Document contributions and assignments, review prior disclosures and prior art, and file selectively for qualifying technical inventions. Stage prosecution and international protection by commercial relevance; retain confidential implementation know-how.

The main failure conditions are explicit: gains disappear against competent controls; discovery/learning/traffic consume the savings; depth or adaptation fails to retain useful information; timing precision/control defeats hardware economics; or customers cannot capture enough value after integration. These determine which product path to pursue and when to change course.

The investable thesis is a staged path from an unusual trainable architecture to a repeatable economic advantage, then to a platform spanning the cloud and physical world. We can earn substantial value with a successful first product while retaining the much larger ambition of frontier intelligence and a new computing substrate.

Evidence and commercial diligence

[Evidence manifest](#) records completed-parent hashes, precise metrics, arithmetic conventions and primary market sources accessed on 3 October 2026. [Technical milestones](#), [hardware thesis](#) and [frontier protocol](#) supply the next validation boundaries. No customer commitments, revenue, patent grants, fabricated silicon or current frontier-scale capability are represented as existing assets in this case.